

As X, Do Y: How Persona and Task Combine in Instruction-Tuned LLMs

Eric Xu
Independent Researcher
youxu@wustl.edu

Abstract

Role prompts of the form “*As X, do Y*” admit a clean linear decomposition at one specific site in the residual stream: the prompt-to-answer transition — the last prompt token together with the first two generated tokens — in an early/mid layer band. There, persona and task contribute through partially orthogonal additive directions. Forming a pure persona effect Δ_X , a pure task effect Δ_Y , and substituting $h_{BB} + \Delta_X + \Delta_Y$ for the clean residual yields downstream output within a small KL of clean on Gemma-2-2B-IT and Qwen-2.5-{1.5B, 3B}-Instruct, across a 12-cell short grid and a 48-cell long-persona grid, with persona-specific behavioral markers preserved.

The natural inference from this additive structure is that the role prompt can be compressed into a single cached residual vector. *We show it cannot.* Injecting the cached additive prediction — or even the oracle clean residual h_{XY} — into a baseline host prompt with the persona text removed does not approach the clean long-persona target, at one site or at many layers. Persona-conditioned multi-token generation flows through attention back to the persona-text positions throughout the prompt, which no residual at one site reproduces.

The takeaway is that local additivity in the residual stream does not imply prompt compressibility. Role prompts are two things at once: an additive composition that becomes mechanistically tractable at one small, well-defined region — a natural site for interpretability and for fine-grained steering of persona or task — and a distributed prompt/KV mechanism that produces persona-conditioned behavior across the full continuation and that local activation arithmetic cannot displace.

1 Introduction

Prompts of the form “*As Warren Buffett, give advice*” or “*As a senior software engineer, review this design*” are routine, and instruction-tuned models handle them robustly. The online computation that combines a persona specification X with a task specification Y in the residual stream is less well understood. This paper asks where, along the layer and position axes, that combination is causally simple enough to be approximated additively.

For each persona-task pair we compare four prompts – baseline-baseline (BB), persona-only (XB), task-only (BY), and persona-plus-task (XY) – and at each layer and position form

$$\Delta_X = h_{XB} - h_{BB}, \quad \Delta_Y = h_{BY} - h_{BB}, \quad (1)$$

$$\Delta_{XY} = h_{XY} - h_{BB}, \quad \text{Inter} = \Delta_{XY} - \Delta_X - \Delta_Y. \quad (2)$$

The empirical questions follow directly: is Δ_{XY} aligned with $\Delta_X + \Delta_Y$? Are Δ_X and Δ_Y separated? When we substitute $h_{BB} + \Delta_X + \Delta_Y$ for the clean XY residual, does downstream output stay close to clean? The answers turn out to depend sharply on layer and position. They are affirmative near

answer formation – the last prompt position p_{last} and the first two generated tokens g_1, g_2 , in an early/mid layer band – and they degrade outside that region. The pattern holds across Gemma-2 and Qwen-2.5 instruction-tuned models and across a 48-cell grid of long-form personas and diverse tasks.

The natural reading of an additive decomposition is that the role prompt can be replaced by its cached vector — a single residual injected once should reproduce the long-persona behavior. The same intervention machinery shows that it cannot. Injecting the cached additive prediction at p_{last} into a baseline host prompt with the persona text removed does not approach the clean long-persona target, and clamping the substitution at many layers, or substituting the oracle clean residual h_{XY} itself, does not change this. Multi-token persona-conditioned generation flows through attention back to the persona-text positions throughout the prompt, which no residual at one site reproduces. Local additivity in the residual stream does not imply prompt compressibility; role prompts operate at both scales simultaneously.

2 Related Work

This paper draws on four lines of prior work.

Task and function vectors from contrastive prompts. The closest methodological precedent is the line of work that constructs task representations from differences between contrastive prompts. In-context learning creates task vectors that can be transplanted across prompts [2], and function vectors formalize this as averaged contrastive activation differences that causally drive task-conditioned behavior [8]. Our $\Delta_X, \Delta_Y, \Delta_{XY}$ are residual-stream analogs of this construction, but the focus of this paper is compositional structure (how persona and task combine) rather than single-axis transplantation.

Activation-space steering and representation engineering. Activation Addition [9] and subsequent representation-engineering work [10] show that activation differences can steer model behavior without weight optimization. Task arithmetic [4] demonstrates that fine-tuning deltas behave additively in weight space. Our paper is adjacent in method but different in aim: we are not proposing a steering tool, we use activation substitution to characterize the compositional structure induced by role prompts.

Soft prompts and prefix tuning. Prefix-Tuning [6] and Prompt-Tuning [5] show that a small number of learned continuous vectors prepended to a prompt can drive task-conditioned behavior. Our negative result in [section 7](#) is informative against the most aggressive reading of our positive result: full prompt-to-vector replacement is closer to the soft-prompt regime, which uses many positions and is learned rather than constructed by single-shot residual arithmetic.

Mechanistic editing and feature interpretation. ROME-style work [7] ties factual behavior to localized weight-space mechanisms and demonstrates surgical edits; subsequent work extends the linearity picture to relation decoding [3]. Our paper does not reach that level of closure. Instead, it provides a residual-stream-level account of where persona and task composition is causally simple enough to be approximated additively, and where that story stops. Work on monosemanticity and feature extraction [1] suggests that high-level behaviors sometimes decompose into interpretable latent directions; this paper does not attempt a component-level ontology and argues that the first object here is a localized residual computation, not a catalog of components.

3 Setup

3.1 Models

The primary model is `google/gemma-2-2b-it`, run in `float32`. We use two Qwen instruction-tuned models for cross-model confirmation of the localized position result: `Qwen/Qwen2.5-1.5B-Instruct` and `Qwen/Qwen2.5-3B-Instruct`, both run in `bfloat16`. The choice of `bfloat16` for the Qwen models is load-bearing: in `float16`, residual magnitudes overflow at the layers relevant here and decomposition statistics become NaN. `bfloat16` preserves the dynamic range. All measurements are causal-intervention experiments on hidden states, not weight edits. Generation is fully greedy throughout (`do_sample=False`, `num_beams=1`) so that conditions can be compared directly without averaging over sampling noise.

3.2 Prompt grids

We use two prompt families.

Short grid. The first grid contains 4 personas and 3 tasks:

- Personas: Warren Buffett, Karl Marx, Yoda, Maya Angelou
- Tasks: policy commentary on UBI, haiku about Monday mornings, book recommendation

With baseline persona “*a thoughtful person*” and baseline task “*Give advice to someone facing a difficult decision*”, this yields 12 non-baseline (X, Y) cells.

Long-persona grid. To test broader prompt diversity, we use 8 multi-sentence personas and 6 tasks:

- Personas: engineer, counselor, founder, teacher, journalist, doctor, lawyer, chef
- Tasks: architecture review, startup launch plan review, scheduling proposal review, UBI policy commentary, haiku, book recommendation

This gives 48 non-baseline (X, Y) cells.

3.3 Residual decomposition

For each (X, Y) pair, we build the usual 2×2 prompt set:

$$BB, XB, BY, XY.$$

At a chosen layer and sequence position, the four hidden states induce Δ_X , Δ_Y , Δ_{XY} , and Inter as defined above.

We report three families of statistics:

- directional alignment $\cos(\Delta_{XY}, \Delta_X + \Delta_Y)$,
- subspace overlap $\cos(\Delta_X, \Delta_Y)$,
- interaction magnitude $\|\text{Inter}\|/\|\Delta_{XY}\|$.

Model	Position	Early layer	Mid layer	Later layer
Gemma-2-2B-IT	p_{last}	L6: 0.0004 [.000,.001]	L10: 0.033 [.022,.060]	L14: 0.371 [.275,.443]
Gemma-2-2B-IT	g_1	L6: 0.002 [.001,.004]	L10: 0.005 [.003,.033]	L14: 0.049 [.021,.124]
Gemma-2-2B-IT	g_2	L6: 0.001 [.000,.003]	L10: 0.013 [.004,.023]	L14: 0.060 [.024,.139]
Qwen-2.5-1.5B	p_{last}	L7: 0.022 [.016,.028]	L11: 0.026 [.019,.033]	L15: 0.065 [.030,.136]
Qwen-2.5-1.5B	g_1	L7: 0.027 [.012,.069]	L11: 0.050 [.032,.077]	L15: 0.118 [.070,.206]
Qwen-2.5-1.5B	g_2	L7: 0.016 [.008,.028]	L11: 0.023 [.013,.037]	L15: 0.034 [.016,.125]
Qwen-2.5-3B	p_{last}	L9: 0.031 [.013,.060]	L14: 0.030 [.013,.076]	L20: 0.074 [.021,.196]
Qwen-2.5-3B	g_1	L9: 0.021 [.017,.035]	L14: 0.045 [.020,.147]	L20: 0.104 [.025,.225]
Qwen-2.5-3B	g_2	L9: 0.021 [.011,.029]	L14: 0.026 [.023,.055]	L20: 0.040 [.029,.065]

Table 1: Median causal KL under additive substitution on the 12-cell short grid, with 25th–75th percentile range in brackets. The additive regime is localized to a small region near answer formation rather than confined to a single prompt position.

3.4 Positions

We probe three positions: p_{last} (the last prompt token), g_1 (the first generated token), and g_2 (the second generated token). At g_1 and g_2 we teacher-force on the clean XY continuation, so comparisons across conditions remain position-aligned.

3.5 Causal metric

At each probe position and layer, we substitute the additive prediction

$$h_{BB} + \Delta_X + \Delta_Y$$

for the clean XY hidden state and measure the downstream effect by 10-token teacher-forced KL against the clean XY continuation. Low KL means the additive approximation is causally sufficient for near-term downstream behavior at that site.

4 Localized Additive Composition Near Answer Formation

We measure the 2×2 decomposition at three probe positions – p_{last} , g_1 , g_2 – on the 12-cell short grid. Table 1 and figure 1 summarize the result.

The pattern is the same across positions and models: a low-KL early/mid layer band, smoothly degrading later. On Gemma-2-2B-IT at p_{last} , median causal KL is 0.0004 at layer 6, 0.033 at layer 10, 0.371 at layer 14, 1.371 at layer 18. At g_1 the corresponding numbers are 0.002, 0.005, 0.049; at g_2 , 0.001, 0.013, 0.060. The cross-model picture matches: at the best early layer, Qwen-2.5-1.5B has medians 0.030/0.018/0.014 at $p_{\text{last}}/g_1/g_2$, and Qwen-2.5-3B has 0.031/0.021/0.021. The first generated tokens stay low-KL further into the network than p_{last} does, which is consistent with the additive composition being most legible exactly at the prompt-to-answer transition.

The geometry tells the same story. At Gemma-2-2B-IT layer 14, median $\cos(\Delta_{XY}, \Delta_X + \Delta_Y)$ is 0.874 at p_{last} , 0.935 at g_1 , and 0.898 at g_2 , while $\cos(\Delta_X, \Delta_Y)$ remains substantially lower. Δ_X and Δ_Y are partially separable; the interaction term is non-negligible in norm but does not rotate Δ_{XY} away from the additive sum.

The additive approximation thus identifies a region, not a point: an early/mid layer band, at p_{last} and the first two generated tokens. The boundaries of this region – in particular how it varies with model scale and depth – become the natural object for the rest of the paper.

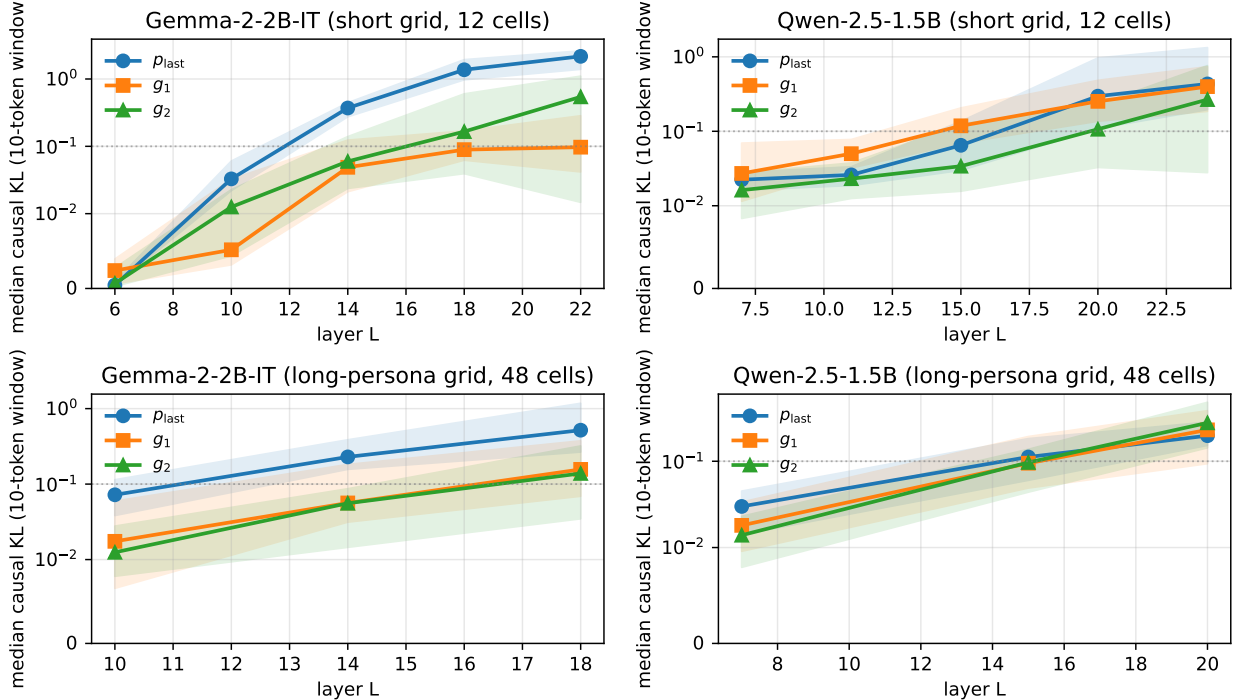


Figure 1: Median causal KL under additive substitution at p_{last} (blue), g_1 (orange), and g_2 (green), with 25th–75th percentile bands. Top row: 12-cell short grid on Gemma-2-2B-IT and Qwen-2.5-1.5B-Instruct. Bottom row: 48-cell long-persona grid on the same two models. Dotted gray line marks $\text{KL} = 0.1$, a useful reference for “essentially clean” substitution. The localized additive regime is visible as an early/mid-layer band with low KL across all three positions, present on both models and on both grids.

5 Robustness to Persona and Task Diversity

The 12-cell grid is convenient but small. We rerun the same localized analysis on an 8-persona $\times$ 6-task grid (48 cells) that includes multi-sentence personas (engineer, counselor, founder, teacher, journalist, doctor, lawyer, chef) and a mix of review, planning, policy, creative, and recommendation tasks (section A).

The localized additive regime persists (table 2; bottom row of figure 1). On Gemma-2-2B-IT, median causal KL at layer 14 is 0.229 at p_{last} , 0.056 at g_1 , and 0.056 at g_2 ; the corresponding cosines $\cos(\Delta_{XY}, \Delta_X + \Delta_Y)$ are 0.818 and 0.875 at p_{last} and g_1 , while $\cos(\Delta_X, \Delta_Y)$ stays substantially lower (0.171, 0.234), ruling out trivial colinearity. On Qwen-2.5-1.5B at early layer 7, the medians are 0.030/0.018/0.014, with cosines 0.862 and 0.936. Quantitatively the long-persona numbers are larger than the short-grid numbers – multi-sentence personas push more into the residual – but the qualitative shape is unchanged.

Per-cell breakdown. The effect is not uniform across cells, but it is not driven by a few easy ones either. Per-persona median KLs on Gemma-2-2B-IT at p_{last} , layer 14 range from 0.14 to 0.39 for most personas, with a few outlier cells in the tail; on Qwen-1.5B-Instruct at layer 7 the persona medians are tighter, mostly 0.02–0.035. Per-task medians on Gemma remain low to moderate across all six task families rather than concentrating on one.

Model	Position	Early layer	Mid layer	Later layer
Gemma-2-2B-IT	p_{last}	L10: 0.072 [.038,.114]	L14: 0.229 [.157,.386]	L18: 0.518 [.266,1.17]
Gemma-2-2B-IT	g_1	L10: 0.017 [.007,.059]	L14: 0.056 [.031,.182]	L18: 0.157 [.069,.374]
Gemma-2-2B-IT	g_2	L10: 0.012 [.008,.028]	L14: 0.056 [.015,.087]	L18: 0.138 [.035,.320]
Qwen-2.5-1.5B	p_{last}	L7: 0.030 [.014,.045]	L15: 0.112 [.060,.183]	L20: 0.198 [.153,.284]
Qwen-2.5-1.5B	g_1	L7: 0.018 [.010,.034]	L15: 0.095 [.049,.193]	L20: 0.231 [.094,.385]
Qwen-2.5-1.5B	g_2	L7: 0.014 [.008,.023]	L15: 0.097 [.044,.137]	L20: 0.279 [.143,.481]

Table 2: Median causal KL on the 48-cell long-persona grid, with 25th–75th percentile range in brackets. The localized additive regime remains visible under substantially broader persona and task diversity.

6 Behavioral Verification of the Additive Substitution

KL between teacher-forced distributions is a useful but indirect measure of whether the additive substitution preserves persona-conditioned behavior. A persona prompt should change *what* the model talks about, not only the local probability of the next few tokens. We therefore complement the KL evidence in [sections 4](#) and [5](#) with a behavioral-marker test that scores persona-specific output content directly, in the style of factual recall verification [\[7, 8\]](#).

6.1 Protocol

For each of the 8 long-form personas we define a small set of persona-specific surface markers chosen *a priori* from the persona description itself. For example, the engineer persona is associated with markers including `SPOF`, `single point of failure`, `scalability`, `reliability`, `redundancy`; the counselor persona with markers including `feel heard`, `validate`, `compassion`; the doctor persona with `differential`, `symptoms`, `ruling out`. The full marker sets are listed in [section B](#). We hold out the marker selection from the experiment design — markers are picked from the persona text, not from clean outputs.

For each (long persona, task) cell on the 8-persona $\times$ 3 review-task subset of the long-persona grid (24 cells), we generate 80 tokens greedily under four conditions:

- **Clean:** “*As [long persona], [task]*” with no intervention. This is the ceiling.
- **Additive:** same prompt, with $h_{BB} + \Delta_X + \Delta_Y$ substituted at p_{last} , layer 14.
- **Remove-X:** same prompt, with $h_{XY} - \Delta_X$ substituted at p_{last} , layer 14 (subtract the persona contribution).
- **Bare:** “*[task]*” with no persona prefix and no intervention. This is the floor.

A cell scores *any-marker* if at least one persona marker appears in the 80-token output (case-insensitive, word-boundary-aware). A faithful additive substitution should match the clean condition on this metric while remove-X and bare both drop substantially below it.

6.2 Results

The bare condition collapses to 4.2%, confirming that the persona text is the primary source of persona-conditioned content in this protocol. The additive condition tracks the clean ceiling closely

Condition	Any persona marker present	Distinct markers (mean)
Clean	16 / 24 (66.7%)	1.21
Additive substitution at p_{last} , $L = 14$	14 / 24 (58.3%)	0.92
Remove-X at p_{last} , $L = 14$	15 / 24 (62.5%)	0.96
Bare prompt (no persona text)	1 / 24 (4.2%)	0.08

Table 3: Behavioral-marker recovery on 8-persona $\times$ 3-task long-persona cells, scoring 80-token greedy continuations. “Any” columns count cells in which at least one persona marker appears; “distinct” columns count distinct markers per generation.

Condition	Median KL to clean long-persona target
Host prompt, no substitution	3.05
Oracle single-site substitution at L14	2.64
Cached additive substitution at L14	2.84
Cached substitution at $\{10, 12, 14\}$	2.81
Cached substitution at $\{10, 12, 14, 16, 18\}$	2.82
Cached substitution at $\{10, 12, 14, 16, 18, 20, 22\}$	3.04
Cached substitution at $\{6, 8, 10, 12, 14, 16, 18, 20, 22\}$	3.01

Table 4: Long-persona host-prompt injection does not approach clean long-persona behavior. Better local additive prediction does not imply prompt replacement.

(58.3% vs. 66.7%; 0.92 vs. 1.21 distinct markers), so substituting $h_{BB} + \Delta_X + \Delta_Y$ at p_{last} , $L = 14$ does not damage the persona-conditioned output.

Remove-X is the informative third row. Subtracting Δ_X at one site barely changes marker presence (62.5%). This is consistent with the negative result in [section 7](#): during multi-token generation, persona content arrives via attention back to the persona-text positions, not through a single residual at p_{last} . A one-site subtraction cannot reach those positions. Only removing the persona text from the prompt itself – the bare condition – collapses persona-marker recovery.

7 Where the Local Additive Story Stops

A natural next question is whether the persona prompt can be replaced by the cached residual vector itself. It cannot. We construct, for each long-persona cell, a host prompt with the persona text removed (“*As a thoughtful person, [task]*”) and compare 10-token KL against the clean long-persona continuation under three interventions at p_{last} : no substitution, oracle substitution with the clean h_{XY} , and cached substitution with $h_{BB} + \Delta_X + \Delta_Y$. Across 24 cells, host-baseline median KL is 3.05, cached substitution reaches 2.84, oracle substitution reaches 2.64. Both close only a small fraction of the gap.

Multi-layer substitution does not help. Clamping the additive prediction at successively wider layer sets gives medians 2.81, 2.82, 3.04, 3.01 for layer sets $\{10, 12, 14\}$, $\{10, 12, 14, 16, 18\}$, $\{10, 12, 14, 16, 18, 20, 22\}$, $\{6, 8, 10, 12, 14, 16, 18, 20, 22\}$. Wider windows eventually hurt, because clamping over many layers prevents the natural forward propagation that would otherwise carry persona content through downstream computation.

The picture is two-level. The local additive regime characterizes the next-step composition: at p_{last} and the first two generated tokens, in an early/mid layer band, $\Delta_X + \Delta_Y$ is causally sufficient. Multi-token persona-conditioned generation depends on attention back to the persona-text positions across the prompt, which a single residual cannot reproduce regardless of how many layers it is

clamped over.

8 Discussion

The natural mechanistic object for online role prompting is not a single hidden state but a small region: p_{last} together with the first few generated tokens, in an early/mid layer band. Within this region, Δ_X and Δ_Y are reasonable summaries of the persona and task contributions, and their sum is a useful predictor of the composite residual. The structure survives both a 48-cell long-persona grid and cross-model checks on Gemma and Qwen, which gives the localization some claim to generality.

Outside the region the story degrades, and the host-injection experiment in [section 7](#) sets the outer boundary. Persona-conditioned generation reaches the output through attention to persona-text positions throughout the prompt, and a residual at one site cannot stand in for that. The picture is two-level: local additive composition at the prompt-to-answer transition, and a wider distributed prompt/KV mechanism that local substitution does not displace.

Relative to function-vector [8] and task-vector [2] work, Δ_X and Δ_Y are residual-stream analogs of the same construction; the contribution here is compositional rather than single-axis. Relative to activation-addition steering [9, 10] the aim is characterization, not control. Relative to ROME-style editing [7], the analysis sits at a higher level of abstraction – residual composition rather than weight-space facts – and does not claim circuit-level closure.

On the interaction term. The interaction can be large in raw residual norm even where additive substitution remains causally strong. One candidate explanation is that the residual consumed by the next layer is post-normalization, and the next-layer RMSNorm absorbs part of the apparent magnitude gap between $h_{BB} + \Delta_X + \Delta_Y$ and h_{XY} before downstream computation sees it. We do not directly intervene on the normalization path here, so we report this only as a candidate. The empirical observation, which does not depend on it, is that raw non-additivity in norm does not predict downstream causal fidelity under substitution.

On position choice. The region $(p_{\text{last}}, g_1, g_2)$ spans the transition from prompt representation to the first answer tokens. Earlier positions inside the user turn can also show low-KL additive substitution, but with less stable geometry. Later generated positions drift increasingly under teacher forcing.

9 Limitations

1. The analysis is restricted to a small region near answer formation. We do not study every prompt position or every generated token.
2. We do not identify a minimal set of heads, neurons, or weights whose editing installs or removes a persona feature.
3. The long-persona grid is broader than the short grid but remains English-only, cooperative, and single-turn.
4. Some late-layer and particular-cell tails are large. We do not characterize which (X, Y) combinations push the interaction term beyond the regime where additive substitution remains causally faithful.

5. The host-injection experiment shows that single-site substitution cannot replace persona text, but does not identify the correct distributed intervention.
6. Cross-model evidence is limited to two families. The 12-cell short-grid localized result was checked on Gemma-2-2B-IT and Qwen-2.5-{1.5B, 3B}; the 48-cell long-persona grid was run on Gemma-2-2B-IT and Qwen-2.5-1.5B-Instruct only.

10 Conclusion

Persona-task composition in instruction-tuned language models admits a localized additive description in the residual stream near answer formation, holds across Gemma-2-2B-IT and Qwen-2.5-{1.5B, 3B}-Instruct on a 48-cell long-persona grid, and survives a behavioral test of persona-conditioned output content. The natural reading of this – that the role prompt can be replaced by the cached vector – is wrong, and the same intervention machinery shows why: persona-conditioned multi-token generation depends on attention back to the persona-text positions, and no residual at one site can reproduce that. Local additivity in the residual stream does not imply prompt compressibility. The natural next step is to characterize the distributed prompt/KV mechanism directly, and to understand the precise interface between it and the local additive regime.

Reproducibility. Code, cached experiment outputs, and the paper source are available at <https://github.com/xuy/localized-additive-composition>. All experiments use HuggingFace `transformers` on a single Apple Silicon device with the `mps` backend. Gemma-2-2B-IT is loaded in `float32`; Qwen-2.5-1.5B-Instruct and Qwen-2.5-3B-Instruct are loaded in `bfloat16` (`float16` produces NaN residuals at the layers relevant here). Attention uses the `eager` implementation. All generation is greedy (`do_sample=False`, `num_beams=1`), so reported KLs and behavioral scores are deterministic functions of the model, prompt, and intervention. KL is computed by greedy 10-token continuation from the clean *XY* prompt, followed by teacher-forced log-probability comparison across the same 10-token reference window under each intervention.

Code and artifacts. The main scripts supporting the paper are:

- `scripts/v15_localized_positions.py` – localized-position causal sweep (section 4)
- `scripts/v16_diverse_grid.py` – broadened 8×6 long-persona grid (section 5)
- `scripts/v17_behavioral_markers.py` – behavioral-marker recovery test (section 6)
- `scripts/v14b_v2_inject.py` – host-prompt single-site substitution (section 7)
- `scripts/v14e_multilayer_inject.py` – multi-layer substitution (section 7)

A Prompt Grids

A.1 Original short grid

Personas. The short-grid persona set is:

- Warren Buffett
- Karl Marx

- Yoda
- Maya Angelou

Tasks. The short-grid task set is:

- “Comment on whether universal basic income is a good policy.”
- “Write a haiku about Monday mornings.”
- “Recommend a book worth reading and explain why.”

Baselines. The baseline persona is “a thoughtful person” and the baseline task is “Give advice to someone facing a difficult decision.”

A.2 Broadened long-persona grid

Personas. The long-persona set is:

- a senior software engineer with 10 years of experience who pays close attention to architecture, reliability, and avoiding single points of failure
- an empathetic counselor with deep training in active listening, cognitive behavioral therapy, and trauma-informed care, who helps clients feel heard without imposing solutions
- a pragmatic startup founder who has bootstrapped three companies, makes capital-efficient decisions, iterates fast based on user feedback, and avoids vanity metrics
- a middle school science teacher who has taught for 15 years, explains concepts with relatable analogies, gently checks for understanding, and meets students at their level
- an investigative journalist who has covered city government for two decades, asks pointed questions, follows the money, and verifies every claim against primary sources
- a primary-care physician who has practiced for 25 years, listens carefully to symptoms, considers differential diagnoses without alarming the patient, and explains options clearly
- a corporate litigator who has tried cases at the appellate level for 20 years, anticipates opposing arguments, builds case theory from the record, and communicates dense law in plain English
- a head chef trained in classical French technique who has run three Michelin-starred kitchens, builds menus around seasonal ingredients, and teaches young cooks by demonstration

Tasks. The long-persona task set is:

- architecture review: “Review this design: a microservice architecture where eight services share a single PostgreSQL database for both transactional state and event log.”
- startup-plan review: “Review this plan: a three-person team building a B2B SaaS product, planning to launch in three months, with no usage analytics in v1.”
- scheduling-proposal review: “Review this proposal: an internal tool that automates calendar scheduling using an LLM, sending tentative meetings to all parties before confirmation.”

- policy commentary: “Comment on whether universal basic income is a good policy.”
- constrained creative: “Write a haiku about Monday mornings.”
- recommendation: “Recommend a book worth reading and explain why.”

B Behavioral Markers

The persona marker sets used in [section 6](#). Markers were selected from each persona’s description before running the experiment.

- **engineer:** SPOF, single point of failure, scalability, scalable, reliability, fault tolerance, fault-tolerant, redundancy, resilience, throughput, latency, consistency, availability.
- **counselor:** feel heard, validate, validated, acknowledge, acknowledged, your experience, your feelings, compassion, compassionate, without judgment, trauma-informed, active listening, feelings, emotion.
- **founder:** iterate, iterating, iteration, MVP, minimum viable, user feedback, customer feedback, capital-efficient, lean, runway, validation, ship, shipping, product-market fit, vanity metric, traction, burn, bootstrapped.
- **teacher:** imagine, think of, like a, analogy, analogies, for example, step by step, step-by-step, students, understand, let’s say, picture this, as if.
- **journalist:** sources, source, primary source, primary sources, follow the money, accountability, transparency, investigate, verify, verified, on the record, off the record, evidence, public interest, pointed question, track record.
- **doctor:** symptom, symptoms, diagnosis, diagnose, differential, patient, treatment, ruling out, rule out, clinical, examination, condition, medication, evaluate, underlying, comorbid.
- **lawyer:** liability, liabilities, jurisdiction, statute, precedent, evidence, evidentiary, opposing, counterparty, due diligence, parties, indemnify*, contractual, compliance, compliant, jurisprudence, case theory, on the record.
- **chef:** seasonal, season, ingredient, ingredients, flavor, flavour, palate, fresh, simmer, sauté, saute, balance, garnish, mise en place, technique, classical, French.

C Artifact Map

The key result files used in the paper are:

- results/v15_gemma2b_localized_positions.json
- results/v15_qwen15_localized_positions.json
- results/v15_qwen3b_localized_positions.json
- results/v16_gemma2b_diverse_grid.json

- `results/v16_qwen15_diverse_grid.json`
- `results/v17_behavioral_markers.json`
- `results/v14b_v2.json`
- `results/v14e_multilayer.json`

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